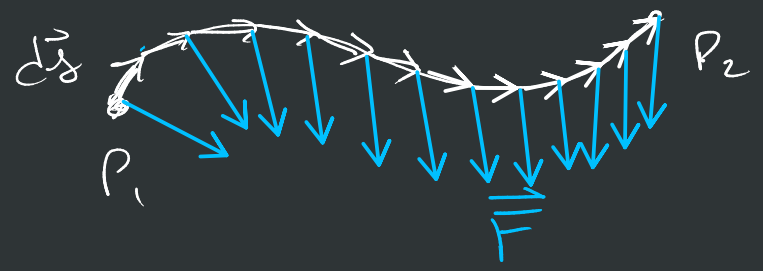
Chapter 2

$$W_{ME} = -\int_{P_1}^{P_2} \vec{F} \cdot d\vec{s}$$



$$\vec{F} = q\vec{E}$$

$$W_{ME} = -\int_{P_1}^{P_2} q\vec{E} \cdot d\vec{s}$$

$$\frac{W_{ME}}{q} = -\int_{P_1}^{P_2} \vec{E} \cdot d\vec{s}$$

$\Delta U = W_{ME}$

$\Delta U = -W_{\vec{E}}$

$\Delta K = 0$

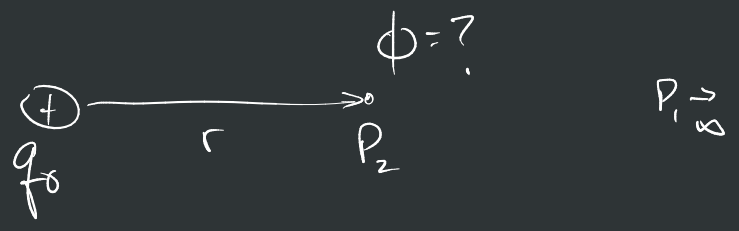
book notation

$$\frac{U(P_2) - U(P_1)}{q} = \frac{\Delta U}{q} = \Delta\phi = \phi_{21} = -\int_{P_1}^{P_1} \vec{E} \cdot d\vec{s}$$

electric potential difference (my notation)

$\downarrow$

$\phi(P_2) - \phi(P_1)$



$$\frac{U}{q} = \phi \qquad \left[\frac{\text{Joule}}{\text{Coulomb}}\right] = [\text{Volt}]$$

Along a closed path $W_{ME} = 0$

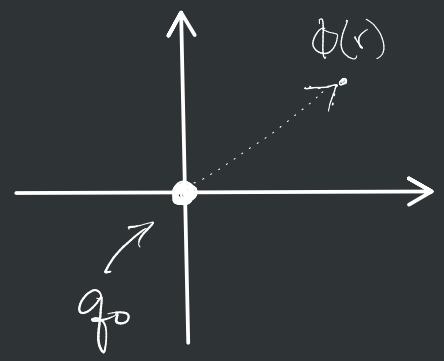
$\Delta U = 0$

$\Delta \phi = 0$

$$\oint \vec{E} \cdot d\vec{s} = 0$$

Ex: $\phi(r)$ for a point charge located at the origin.

$$\phi = -\int_{P_1 \leftarrow \infty}^{P_2 \leftarrow r} \vec{E} \cdot d\vec{s}$$



$$\phi = -\int_{\infty}^{r} \frac{kq_0}{r^2}\hat{r}\cdot dr\,\hat{r}$$

$$\phi = -kq_0 \underbrace{\int_{-\infty}^{r} r'^{-2}\,dr'}_{(-1)r'^{-1} = -\frac{1}{r'}\Big|_{\infty}^{r} = -\frac{1}{r} + \frac{1}{\infty}}$$

$$= -\frac{1}{r}$$

$$\phi(r) = +\frac{kq_0}{r}$$

$$\phi(r) = \frac{q_0}{4\pi\epsilon_0 r}$$

$$U(r) = \frac{q_0 q_1}{4\pi\epsilon_0 r} \qquad \phi = \frac{U}{q_1} = \frac{q_0}{4\pi\epsilon_0 r}$$

✓

<u>Ex:</u> Find ϕ inside + outside a uniformly charged sphere.

ρ

<u>outside</u> $r > R$

$$\phi = -\int_{\infty}^{r} \vec{E}\cdot d\vec{r}$$

$$E = \frac{\rho R^3}{3\epsilon_0 r^2} = \frac{Q}{4\pi\epsilon_0 r^2} \qquad \underline{Q = \rho\,\tfrac{4}{3}\pi R^3}$$

$$\phi(r)_{\text{outside}} = \frac{Q}{4\pi\epsilon_0 r} = \frac{\rho R^3}{3\epsilon_0 r}$$

inside $r < R$

$$E = \frac{\rho r}{3\epsilon_0} \qquad \frac{3Qr}{4 \cdot 3\epsilon_0 \pi R^3} = \frac{Qr}{4\pi\epsilon_0 R^3}$$

$$\phi = -\int \frac{\rho r}{3\epsilon_0} \cdot dr$$

$$\phi(r) = \underbrace{-\int_{-\infty}^{R} E_{out}\, dr}_{} - \int_{R}^{r} E_{inside} \cdot dr$$

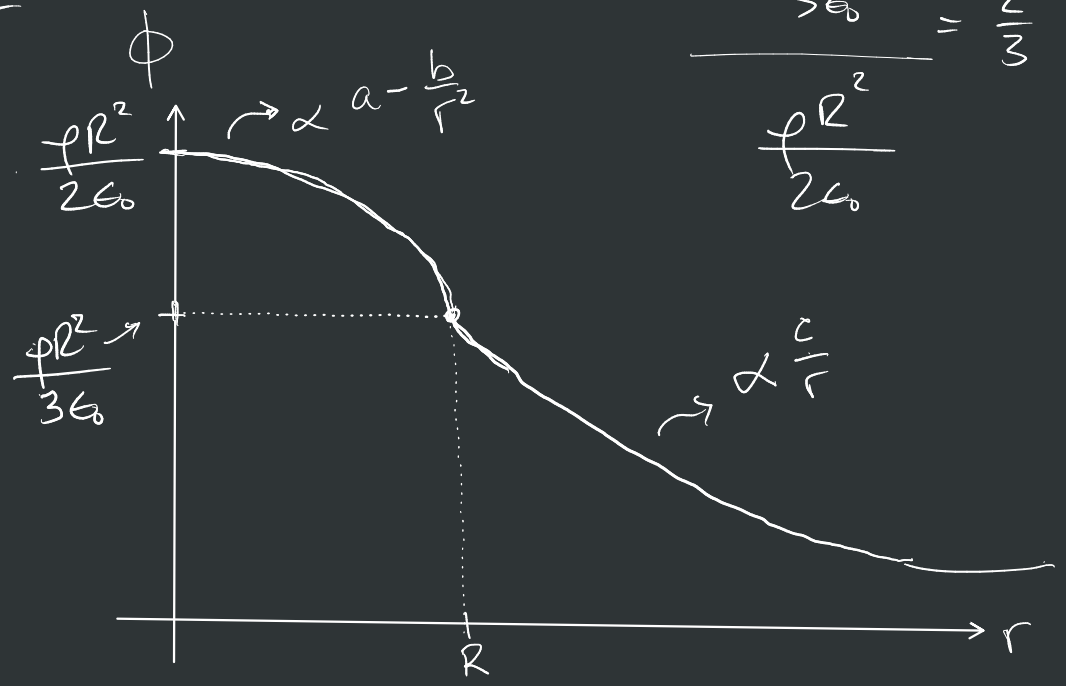
$$\phi(r) = \frac{\rho R^{\cancel{3}\,2}}{3\epsilon_0 \cancel{R}} - \int_{R}^{r} \frac{\rho r'}{3\epsilon_0} dr'$$

$$= \frac{\rho R^2}{3\epsilon_0} - \frac{\rho}{3\epsilon_0} \cdot \frac{1}{2} r^2 \Big|_R^r$$

$$= \frac{\rho R^2}{3\epsilon_0} - \frac{\rho}{6\epsilon_0}(r^2 - R^2)$$

$$\frac{2}{2}\frac{\rho R^2}{3\epsilon_0} + \frac{\rho R^2}{6\epsilon_0} - \frac{\rho r^2}{6\epsilon_0}$$

$$\phi(r)_{inside} = \frac{\rho R^2}{2\epsilon_0} - \frac{\rho r^2}{6\epsilon_0}$$



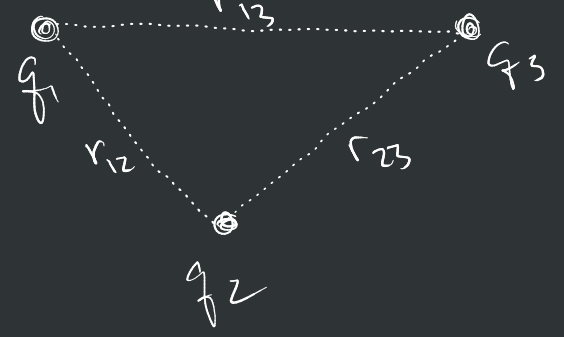
$$\frac{\frac{\rho R^2}{3\epsilon_0}}{\frac{\rho R^2}{2\epsilon_0}} = \frac{2}{3}$$

$$\left(\frac{1}{2} - \frac{1}{6}\right)\frac{\rho R^2}{\epsilon_0}$$

$$\frac{3}{6} - \frac{1}{6}$$

$$\frac{2}{6} = \frac{1}{3}$$

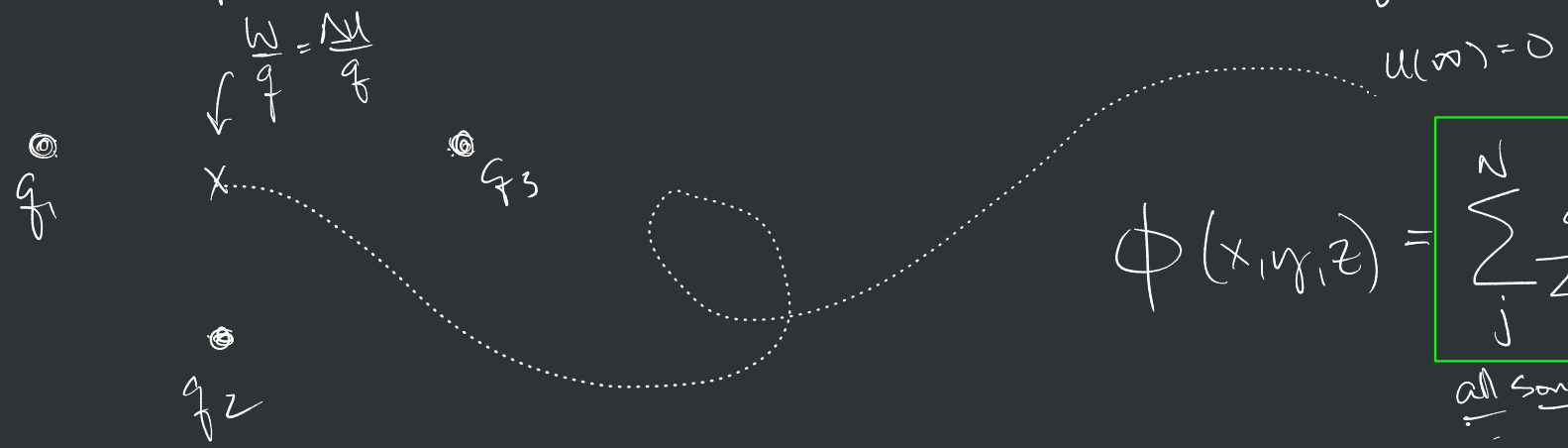
- Potential energy of a collection of point charges:



$$W_{ME} = U = \frac{1}{4\pi\epsilon_0}\left(\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}}\right)$$

$$U = \frac{1}{2}\sum_{j}^{N}\sum_{k\neq j}^{N}\frac{1}{4\pi\epsilon_0}\frac{q_j q_k}{r_{jk}}$$

- <u>Electric potential</u> around a collection of charges:



$$\phi(x,y,z) = \boxed{\sum_{j}^{N}\frac{q_j}{4\pi\epsilon_0 r_j}}$$

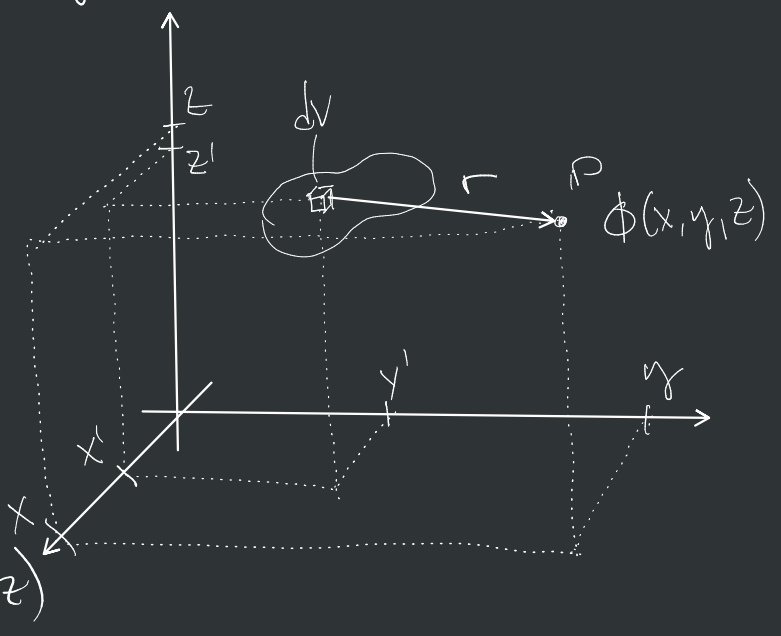
<u>all sources</u>

- Electric potential around a continuous charge distribution

$$q_j \to dq \to \rho\, dV$$

$$\phi(x,y,z) = \int \frac{\rho(x',y',z')\,dx'\,dy'\,dz'}{4\pi\epsilon_0 r}$$

r is distance from (x',y',z') to (x,y,z)



- Relationship between the work (potential energy) to build a continuous distribution and the electric potential of that distribution

$$U = \frac{1}{2}\sum_{j}^{N}\sum_{k\neq j}^{N}\frac{1}{4\pi\epsilon_0}\frac{q_j q_k}{r_{jk}}$$

$$U = \frac{1}{2}\underbrace{\boxed{\sum_{j}^{N} q_j}}_{\rho}\ \underbrace{\boxed{\sum_{k\neq j}\frac{1}{4\pi\epsilon_0}\frac{q_k}{r_{jk}}}}_{\phi}$$

continuous

$$\boxed{U = \frac{1}{2}\int \rho\,\phi\,dv}$$ eq. 2.32

source volume

another way

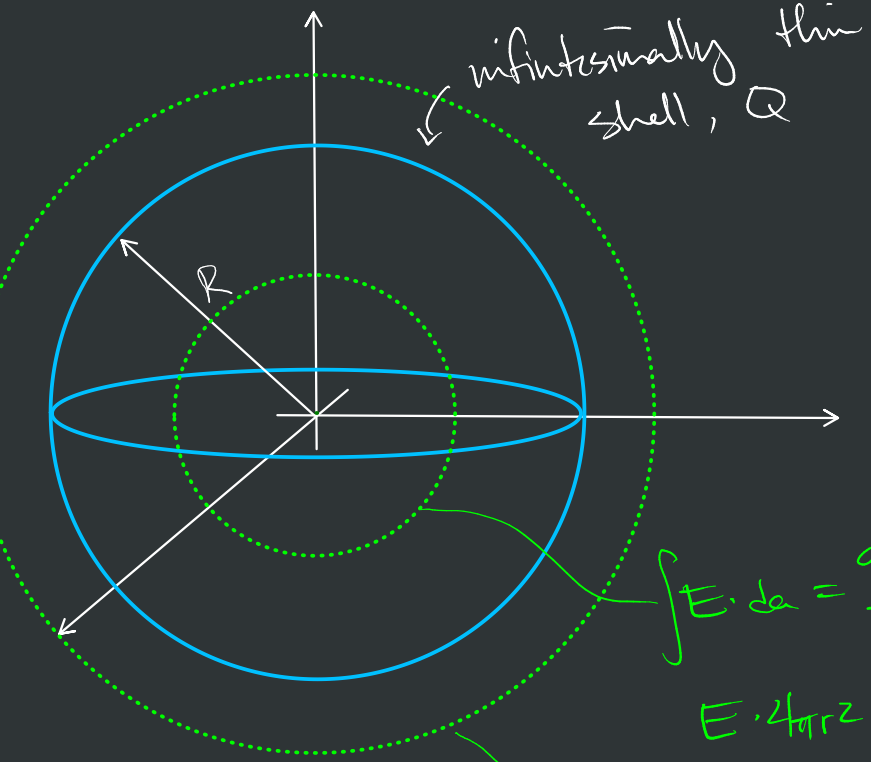
$$U = \frac{\epsilon_0}{2}\int_{\text{entire field}} \vec{E}\cdot\vec{E}\,dv$$

$$\vec{E}\cdot\vec{E} = E^2$$

$$\boxed{U = \frac{\epsilon_0}{2}\int_{\text{entire field}} E^2\,dv}$$

Chapter 1 eq. 1.53

Problem 1.32



$\vec{E} = ?$

$\phi = ?$

$U = ?$

$$\int E \cdot da = \frac{q_{encl}}{\epsilon_0} = 0$$

$$E \cdot 4\pi r^2 = 0$$

$$E = 0$$

inside

$$\vec{E} = \frac{Q}{4\pi\epsilon_0 r^2}\hat{r}$$

outside

$$U = \frac{\epsilon_0}{2}\int \rho\, \phi\, dV$$

$$\rightarrow U = \frac{\epsilon_0}{2}\int_{\text{entire field}} E^2\, dV$$

$$U = \underbrace{\frac{\epsilon_0}{2}\int_{\text{field inside}} E^2\, dV}_{0} + \underbrace{\frac{\epsilon_0}{2}\int_{\text{field outside}} E^2\, dV}$$

$$dV = r^2 \sin\theta\, dr\, d\theta\, d\phi$$

$$U = \frac{\epsilon_0}{2}\int_R^{\infty}\int_0^{\pi}\int_0^{2\pi}\left(\frac{Q}{4\pi\epsilon_0 r^2}\right)^2 r^2 \sin\theta\, dr\, d\theta\, d\phi$$

$$\boxed{\int_0^{2\pi} d\phi = 2\pi}$$

$$\boxed{\int_0^{\pi} \sin\theta\, d\theta = 2}$$

$$U = \frac{\cancel{\epsilon_0}}{2}\left(\frac{Q^2}{4^2\pi^2\epsilon_0^{\cancel{2}}}\right)(\cancel{4\pi})\int_R^\infty \frac{1}{r^2}\,dr$$

$$\int_0^\pi \sin\theta\, d\theta$$

$$-\cos\theta\Big|_0^\pi$$

$$-(-1)-(-(+1))$$

$$+1+1=2$$

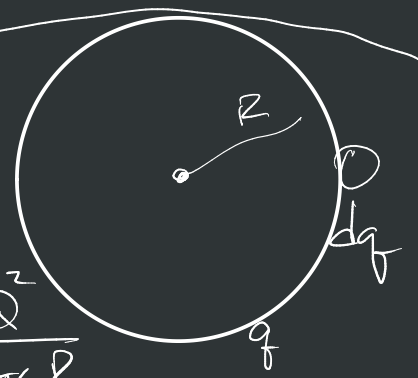
$$U = \frac{Q^2}{8\pi\epsilon_0}(-1)\left(\frac{1}{r}\right)_R^\infty$$

$$U = \frac{-Q^2}{8\pi\epsilon_0}\left[\cancel{\frac{1}{\infty}}^{\,0} - \frac{1}{R}\right]$$

$$\boxed{U = \frac{Q^2}{8\pi\epsilon_0 R}}$$

Find U by adding dq layer at R distance until you have Q.

$$U = \int_0^U dU$$



$$dU = \frac{k\,q\,dq}{R}$$

$$U = \int_0^Q \frac{q\,dq}{4\pi\epsilon_0 R} = \frac{q^2}{8\pi\epsilon_0 R}\Big|_0^Q = \frac{Q^2}{8\pi\epsilon_0 R}$$

Find ϕ.

$$\phi = -\int_{\infty}^{r} \vec{E}\cdot d\vec{r} = -\int_{\infty}^{r} \frac{Q}{4\pi\epsilon_0 r'^2}\,dr' = \frac{Q}{4\pi\epsilon_0 r'}\Big|_{\infty}^{r} = \frac{Q}{4\pi\epsilon_0 r}$$

$$\boxed{\phi(r) = \frac{Q}{4\pi\epsilon_0 r}} \qquad r > R$$

$$\phi(R) = \frac{Q}{4\pi\epsilon_0 R}$$

$$\phi = -\int_{\infty}^{r} \vec{E}\cdot d\vec{r} = -\int_{\infty}^{R} \vec{E}_{r>R}\cdot d\vec{r} - \underbrace{\int_{R}^{r} \overbrace{\vec{E}_{r<R}}^{=0}\, d\vec{r}}_{0}$$

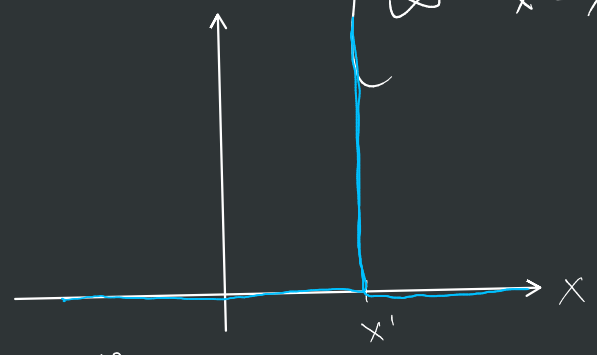
$$= \frac{Q}{4\pi\epsilon_0 R} - 0$$

$$\phi(r) = \frac{Q}{4\pi\epsilon_0 R} \qquad r < R$$

Find $U = \frac{1}{2}\int \rho\,\phi\,dv$.

$\rho = ?$ $\qquad \rho = \dfrac{Q}{V}$ $\quad Q \to \infty,\ V \to 0$ $\longrightarrow$ Dirac Delta

$$\delta(x'-x) = \begin{cases} 0 & x \neq x' \\ \infty & x = x' \end{cases}$$



$$\int_{-\infty}^{\infty} \delta(x'-x)\,dx = 1$$

$$\int_{-\infty}^{\infty} f(x)\,\delta(x'-x)\,dx = f(x')$$

$$Q = \int \rho(r)\,dV$$

$$\boxed{\rho(r) = \rho\,\delta(R-r)}$$

$$Q = \int_{\phi=0}^{2\pi}\int_{\theta=0}^{\pi}\int_{r=0}^{\infty} \rho\,\delta(R-r)\,r^2 \sin\theta\,dr\,d\theta\,d\phi$$

$$Q = 4\pi\rho \int_0^{\infty} r^2\,\delta(R-r)\,dr$$

$$Q = 4\pi R^2 \rho \longrightarrow \rho = \frac{Q}{4\pi R^2}$$

surface charge density

$$U = \frac{1}{2}\int \rho\,\delta(R-r)\,\Phi(r)\,dV$$

$$U = \frac{1}{2}\int_0^{2\pi}\int_0^{\pi}\int_0^{\infty} \rho\,\delta(R-r)\,\Phi(r)\,r^2 \sin\theta\,dr\,d\theta\,d\phi$$

$$U = \frac{1}{2}(4\pi)\int_0^{\infty} \rho\,r^2\,\Phi(r)\,\delta(R-r)\,dr$$

$$U = \frac{1}{2}\cdot(4\pi)\cdot\rho\cdot R^2\,\Phi(R)$$

$$U = \frac{1}{2}(4\pi)\cdot\rho\cdot R^2\cdot\frac{Q}{4\pi\epsilon_0 R}$$

$$\frac{Q}{4\pi R^2}$$

$$U = \frac{1}{2}\cancel{(4\pi)}\,\frac{Q}{\cancel{4\pi R^2}}\cdot\cancel{R^2}\cdot\frac{Q}{4\pi\epsilon_0 R}$$

$$\boxed{U = \frac{Q^2}{8\pi\epsilon_0 R}}$$

$$\phi = -\int_a^b \vec{E} \cdot d\vec{s} \qquad \cancel{\frac{d\phi}{ds} = -E}$$

Go to cartesian coordinates

$$\phi(x, y, z) \longrightarrow \frac{\partial \phi}{\partial x}, \quad \frac{\partial \phi}{\partial y}, \quad \frac{\partial \phi}{\partial z}$$

into a vector

$$\frac{\partial \phi}{\partial x}\hat{x} + \frac{\partial \phi}{\partial y}\hat{y} + \frac{\partial \phi}{\partial z}\hat{z} \longrightarrow$$ points in the direction of <u>steepest</u> <u>change</u>.

This vector is called the <u>gradient</u>

$$\vec{\nabla}\phi = \frac{\partial \phi}{\partial x}\hat{x} + \frac{\partial \phi}{\partial y}\hat{y} + \frac{\partial \phi}{\partial z}\hat{z}$$

$$d\phi = \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy + \frac{\partial \phi}{\partial z} dz$$

$$dy = \frac{dy}{dx} dx$$

also $d\phi = -\vec{E} \cdot d\vec{s}$

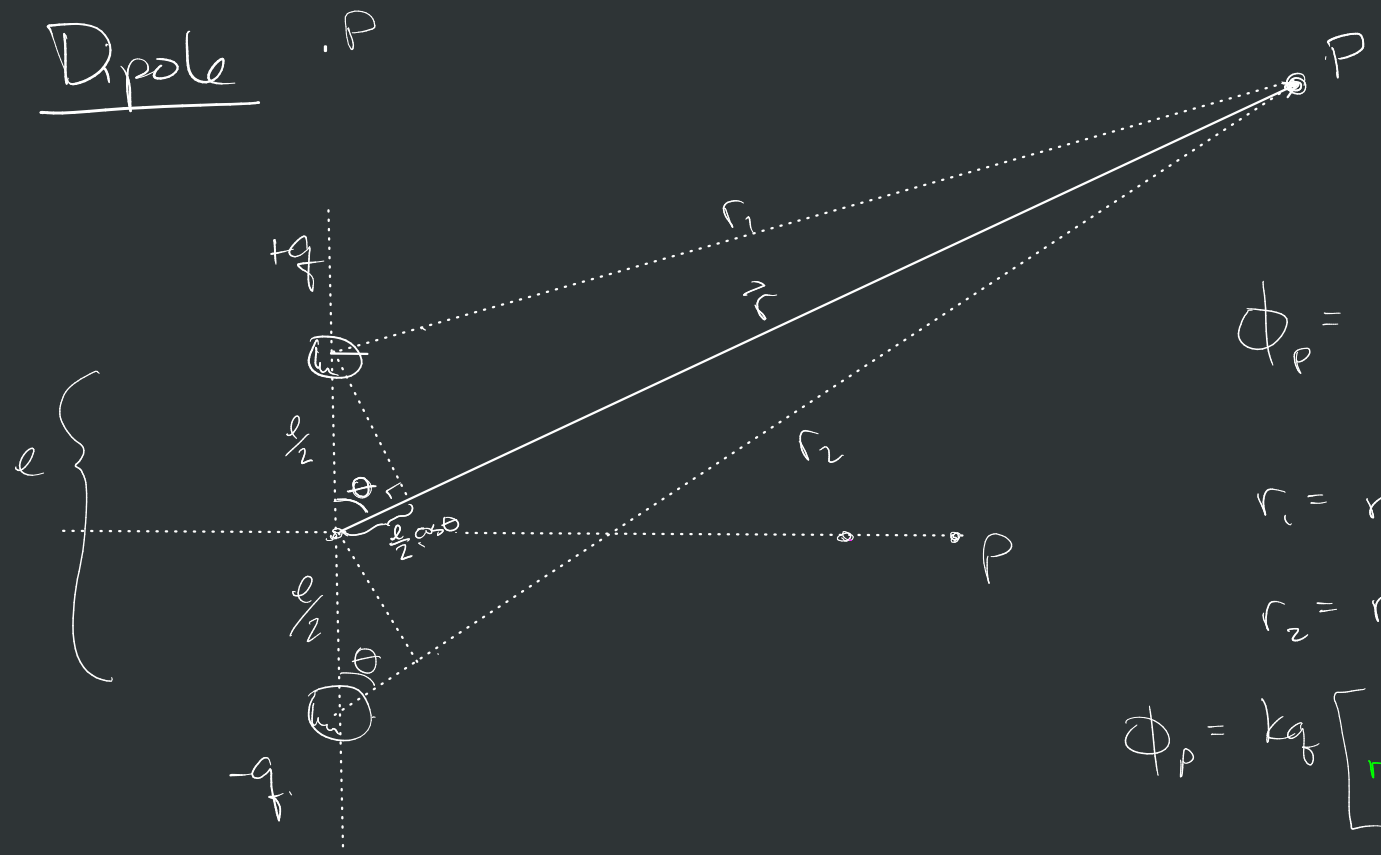
$\hookrightarrow d\vec{s} = dx\,\hat{x} + dy\,\hat{y} + dz\,\hat{z}$

$$d\phi = \underbrace{\frac{\partial \phi}{dx}\hat{x}}\cdot dx\,\hat{x} + \underbrace{\frac{\partial \phi}{\partial y}\hat{y}}\cdot dy\,\hat{y} + \ldots$$

$$d\phi = \vec{\nabla}\phi \cdot d\vec{s}$$

$$\boxed{\vec{E} = -\vec{\nabla}\phi}$$

Dipole



$r \gg \ell$

$$\phi_P = \frac{kq}{r_1} - \frac{kq}{r_2}$$

$$r_1 = r - \frac{\ell}{2}\cos\theta$$

$$r_2 = r + \frac{\ell}{2}\cos\theta$$

$$\phi_P = kq\left[\frac{1}{r\left(\frac{r}{r} - \frac{\ell}{2}\frac{\cos\theta}{r}\right)} - \frac{1}{r\left(\frac{r}{r} + \frac{\ell}{2}\frac{\cos\theta}{r}\right)}\right]$$

$$\phi_P = \frac{kq}{r}\left[\frac{1}{1 - \frac{\ell\cos\theta}{2r}} - \frac{1}{1 + \frac{\ell\cos\theta}{2r}}\right]$$

$$\left(1 - \frac{\ell\cos\theta}{2r}\right)^{-1}$$

$$(1 \pm x)^{-1} \simeq 1 \mp x$$

Taylor expansion to 1st order

$$1 + \frac{\ell\cos\theta}{2r}$$

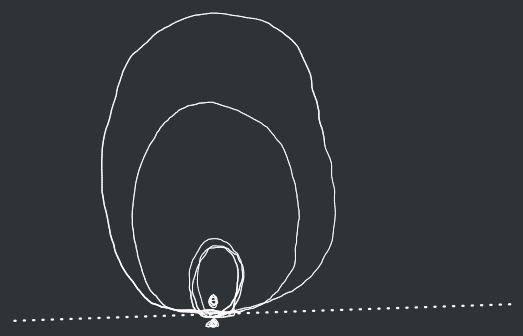
$$\phi_P = \frac{kq}{r}\left[\cancel{1} + \frac{\ell\cos\theta}{2r} - \cancel{1} + \frac{\ell\cos\theta}{2r}\right]$$

$$\phi_p = \frac{kq\,l\cos\theta}{r^2}$$

$ql \equiv p \rightsquigarrow$ <u>dipole</u> <u>moment</u>

$$\longrightarrow \boxed{\phi_p = \frac{kp\cos\theta}{r^2}}$$

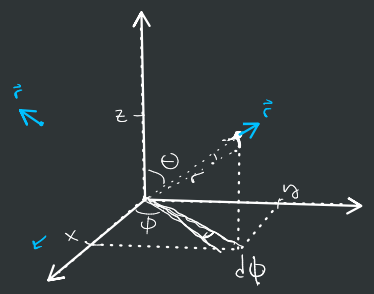
<u>equipotential</u> $\rightsquigarrow$ curve along which the potential is constant.



$$\phi_0 = \frac{kp\cos\theta}{r^2}$$

$$r^2 = \frac{kp}{\phi_0}\cos\theta$$

$$r = A\sqrt{\cos\theta}$$



$E = ?$

$$\vec{\nabla} f = \frac{\partial f}{\partial r}\hat{r} + \frac{1}{r}\frac{\partial f}{\partial \theta}\hat{\theta} + \frac{1}{r\sin\theta}\frac{\partial f}{\partial \varphi}\hat{\varphi}$$

$$E = -\vec{\nabla}\phi = -\vec{\nabla}\left(\frac{kp\cos\theta}{r^2}\right) = -\frac{\partial\left(\frac{kp\cos\theta}{r^2}\right)}{\partial r}\hat{r} - \frac{1}{r}\frac{\partial\left(\frac{kp\cos\theta}{r^2}\right)}{\partial\theta}\hat{\theta} - \frac{1}{r\sin\theta}\underbrace{\frac{\partial\left(\frac{kp\cos\theta}{r^2}\right)}{\partial\phi}}_{0}\hat{\phi}$$

$$= -\left(\frac{-2kp\cos\theta}{r^3}\right)\hat{r} - \frac{1}{r}\left(-\frac{kp\sin\theta}{r^2}\right)\hat{\theta}$$

$$\vec{E} = \frac{2kp\cos\theta}{r^3}\hat{r} + \frac{kp\sin\theta}{r^3}\hat{\theta}$$

$$\vec{E} = \frac{kp}{r^3}\left(2\cos\theta\,\hat{r} + \sin\theta\,\hat{\theta}\right)$$

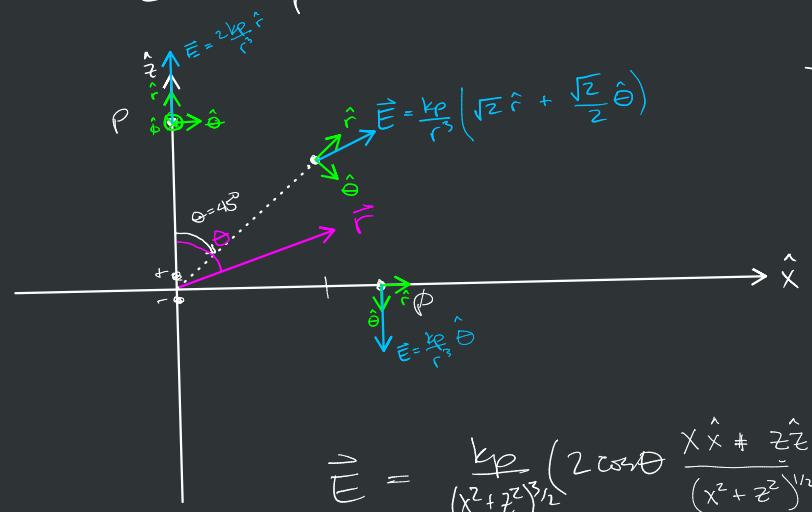
$$\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$$

$$\hat{r} = \frac{x\hat{x} + y\hat{y} + z\hat{z}}{(x^2+y^2+z^2)^{1/2}}$$

$$\hat{\theta} = \cos\theta\cos\phi\,\hat{x} + \cos\theta\sin\phi\,\hat{y} - \sin\theta\,\hat{z}$$

$$\phi = 0$$

$$\hat{\theta} = \cos\theta\,\hat{x} - \sin\theta\,\hat{z}$$



$$\vec{E} = \frac{kp}{(x^2+z^2)^{3/2}}\left(2\cos\theta\,\frac{x\hat{x} + z\hat{z}}{(x^2+z^2)^{1/2}} + \sin\theta\left(\cos\theta\,\hat{x} - \sin\theta\,\hat{z}\right)\right)$$

$$\cos\theta = \frac{z}{r} = \frac{z}{(x^2+z^2)^{1/2}}$$

$$\sin\theta = \frac{x}{r} = \frac{x}{(x^2+z^2)^{1/2}}$$

after some algebra

$$\vec{E} = kp\left(\frac{3xy}{(x^2+y^2)^{5/2}}\hat{x} + \frac{2y^2 - x^2}{(x^2+y^2)^{5/2}}\hat{z}\right)$$

$$\phi_p = \frac{k\rho\cos\theta}{r^2}$$

$$r^2 = x^2 + z^2$$

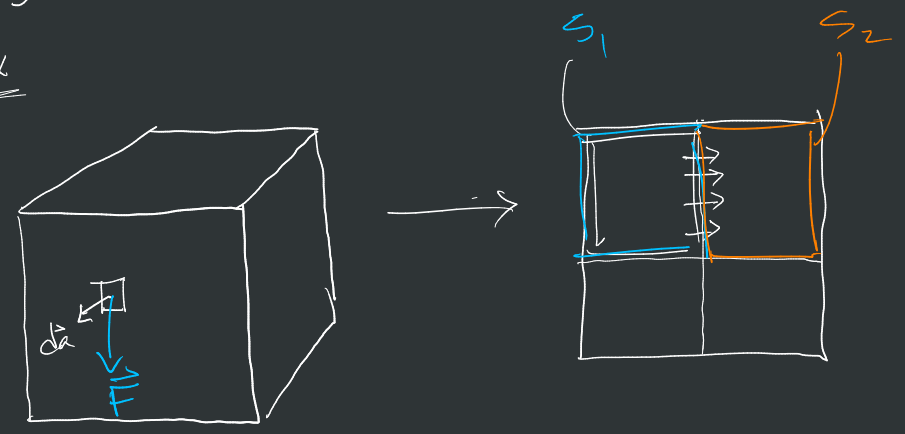
$$\cos\theta = \frac{z}{r} = \frac{z}{(x^2+z^2)^{1/2}}$$

$$\phi_p = k\rho\left(\frac{z}{(x^2+z^2)^{3/2}}\right)$$

<u>Divergence</u> $\rightarrow$ <u>flux</u> <u>density</u>

$$\Phi = \int_S \vec{F}\cdot d\vec{a}$$

<u>flux</u>



$$\Phi = \int_S \vec{F}\cdot d\vec{a}$$

$$\Phi = \int_{S_1} \vec{F}\cdot d\vec{a}_1 + \int_{S_2} \vec{F}\cdot d\vec{a}_2 + \cdots$$

$$\Phi = \int_S \vec{F}\cdot d\vec{a} = \sum_{i=1}^{N} \int_{S_i} \vec{F}\cdot d\vec{a}_i$$

$$\lim_{V_i \to 0} \frac{\int_{S_i} \vec{F} \cdot d\vec{a}_i}{V_i} = \text{div } \vec{F}$$

$$= \vec{\nabla} \cdot \vec{F}$$

operator notation

$$\Phi = \int_S \vec{F} \cdot d\vec{a} = \sum_{i=1}^{N} \frac{V_i}{V_i} \int_{S_i} \vec{F} \cdot d\vec{a}_i$$

now limit as $V_i \to 0$

$$\Phi = \underbrace{\int_S \vec{F} \cdot d\vec{a} = \int_V \vec{\nabla} \cdot \vec{F} \, dv}_{\text{divergence theorem}}$$

$$\vec{\nabla} = \hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z}$$

$$\vec{\nabla} \phi = \left(\hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z} \right) \phi$$

$$\vec{\nabla} \cdot \vec{F} = \left(\hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z} \right) \cdot \left(F_x \hat{x} + F_y \hat{y} + F_z \hat{z} \right)$$

$$\vec{\nabla} \cdot \vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$$

Gauss's Law

$$\underbrace{\int_S \vec{E} \cdot d\vec{a}}_{\text{make a volume integral}} = \frac{1}{\epsilon_0} \int_V \rho \, dV \qquad \frac{q_{encl}}{\epsilon_0}$$

$$\int_S \vec{E} \cdot d\vec{a} = \int_V \boxed{\vec{\nabla} \cdot \vec{E}} \, dv = \int_V \boxed{\frac{\rho}{\epsilon_0}} \, dV$$

$$\boxed{\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}}$$

Gauss's Law in differential form

$$\vec{E} = -\underbrace{\vec{\nabla}\phi}$$

$$\vec{\nabla}\cdot\vec{E} = \frac{\rho}{\epsilon_0}$$

$$\vec{\nabla}\phi = \frac{\partial\phi}{\partial x}\hat{x} + \frac{\partial\phi}{\partial y}\hat{y} + \frac{\partial\phi}{\partial z}\hat{z}$$

$$= \underbrace{\left(\hat{x}\frac{\partial}{\partial x} + \hat{y}\frac{\partial}{\partial y} + \hat{z}\frac{\partial}{\partial z}\right)}_{\vec{\nabla}}\phi$$

$$\vec{\nabla}\cdot\vec{\nabla}\phi = -\frac{\rho}{\epsilon_0}$$

$$\boxed{\nabla^2\phi = -\frac{\rho}{\epsilon_0}}$$ ← Poisson's Equation

↓ special case

$$\boxed{\nabla^2\phi = 0}$$ ← Laplace's Equation

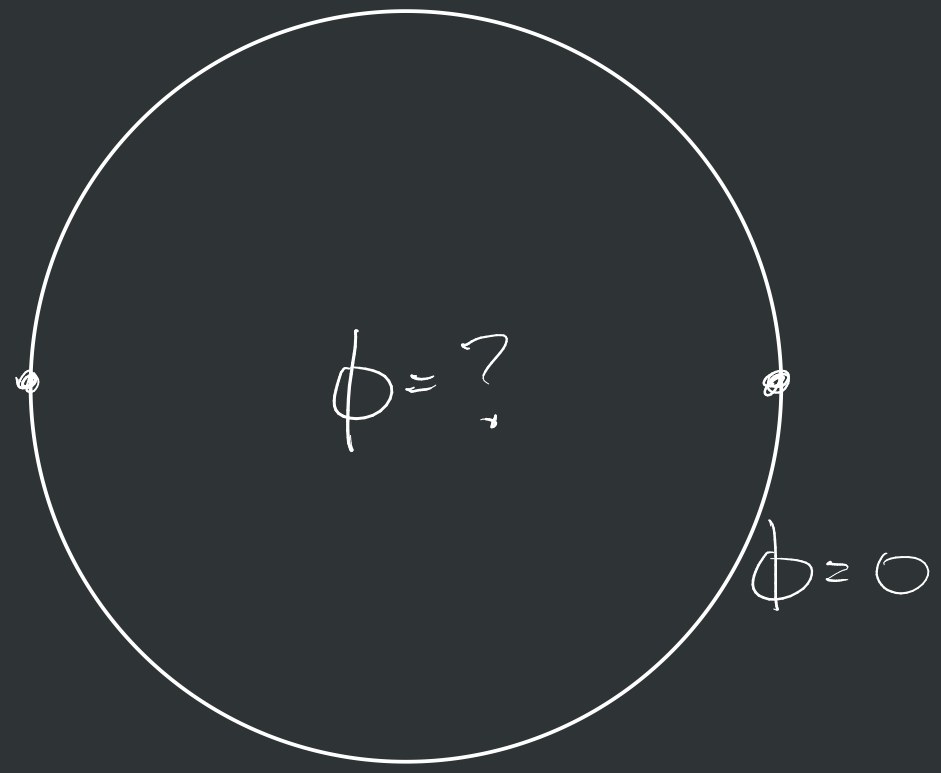
$$\nabla^2 = \vec{\nabla}\cdot\vec{\nabla} = \left(\hat{x}\frac{\partial}{\partial x} + \hat{y}\frac{\partial}{\partial y} + \hat{z}\frac{\partial}{\partial z}\right)\cdot\left(\hat{x}\frac{\partial}{\partial x} + \hat{y}\frac{\partial}{\partial y} + \hat{z}\frac{\partial}{\partial z}\right)$$

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

Laplacian

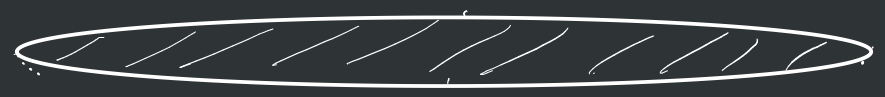
$$\nabla^2\cdot\phi = \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2}$$ ← Cartesian

$$\nabla^2\cdot\phi = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial f}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\phi}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2\phi}{\partial\varphi^2}$$ ← Spherical

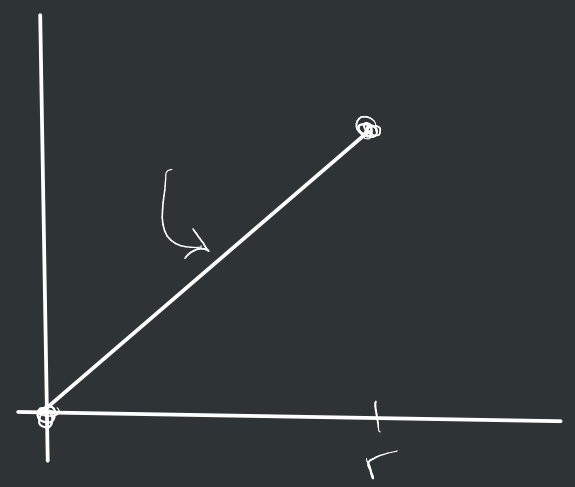


$$\nabla^2 \phi = 0$$

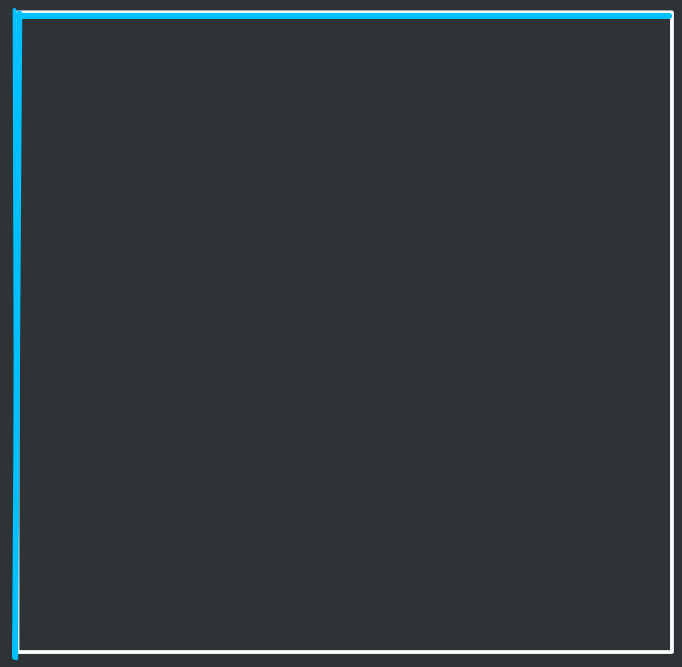
$$F = -\frac{\partial U}{\partial x}$$



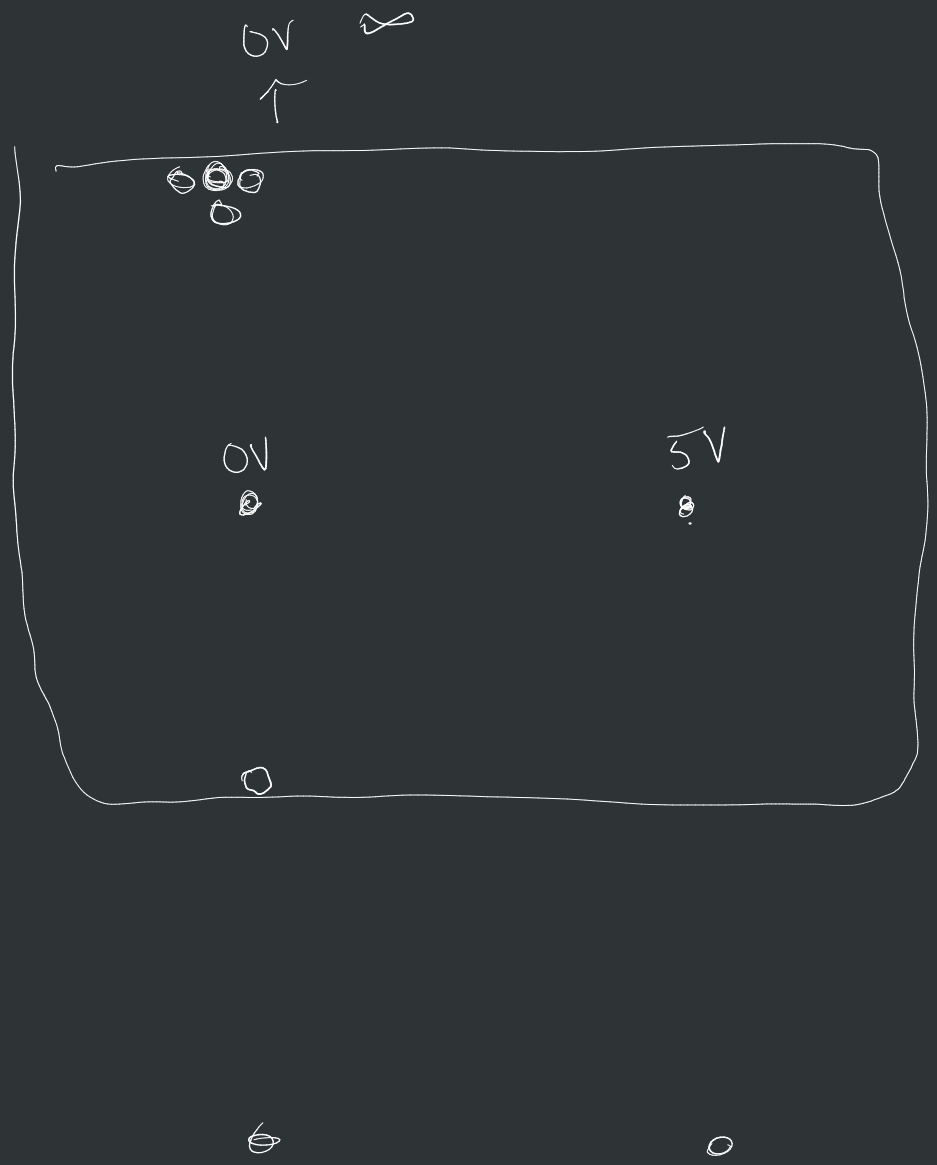
$\phi(r)$



$\phi = \phi_0$

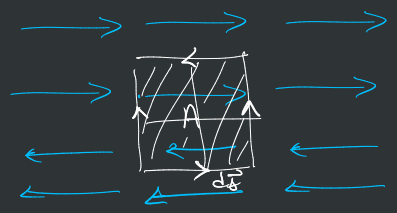


$\phi = 0$



Curl

$$\Gamma = \oint_C \vec{F} \cdot d\vec{s}$$



$$\text{curl}\,\vec{F} = \lim_{a_i \to 0} \frac{\Gamma_i}{a_i} = \lim_{a_i \to 0} \frac{\oint \vec{F} \cdot d\vec{s}}{a_i}$$